

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<think> First let's calculate the net profit the tree generates each year . Each year the tree produces 7 lemons . Each lemon is sold for 1 . 5 dollars . Therefore each year the tree earns $7 * 1 . 5 = 10 . 5$ dollars . Next let's calculate the cost the tree incurs each year . The cost to water and feed the tree is 3 dollars . Additionally the initial planting cost is 90 dollars . However this initial cost is only incurred once . After the initial year the only cost is 3 dollars per year . Now let's calculate the net profit after the first year . After the first year the tree earns 10 . 5 dollars . However the tree also incurs a cost of $90 + 3 = 93$ dollars . Therefore after the first year the tree incurs a loss of $93 - 10 . 5 = 82 . 5$ dollars . Since after the first year the tree incurs a loss . And the tree earns more than it costs after the first year . Therefore the tree will start earning money after the first year . To calculate after how many years . We need to find the break-even point . The break-even point is when the total earnings equal the total costs . Let's assume it takes x years to reach the break-even point . After x years the tree will have earned a total of $10 . 5 x$ dollars . After x years the tree will have incurred a total cost of $93 + 3 x$ dollars . To find the break-even point we set the total earnings equal to the total costs . $10 . 5 x = 93 + 3 x$. Now let's solve for x . First we subtract $3 x$ from both sides . $10 . 5 x - 3 x = 93$. Simplifying the left side we get $7 . 5 x = 93$. Now to isolate x . We divide both sides by 7 . 5 . $x = 93 / 7 . 5 = 12 . 4$. Since x represents the number of years . And it must be a whole number . We round up x to the nearest whole number . Therefore it will take 13 years for the tree to start earning money . </think><answer> 13 </answer>

